US 20250277810A1

(19) **United States**

(12) **Patent Application Publication**

**LIAO et al.**

(10) **Pub. No.: US 2025/0277810 A1**

(43) **Pub. Date: Sep. 4, 2025**

(54) **TESTING APPARATUS AND ITS CONDUCTIVE TERMINAL**

(71) Applicants: **GLOBAL UNICHIP CORPORATION**, HSINCHU CITY (TW); **TAIWAN SEMICONDUCTOR MANUFACTURING COMPANY, LTD.**, Hsinchu (TW)

(72) Inventors: **Chih-Chieh LIAO**, HSINCHU CITY (TW); **Yu-Min SUN**, HSINCHU CITY (TW); **Chih-Feng CHENG**, HSINCHU CITY (TW)

(21) Appl. No.: **18/945,457**

(22) Filed: **Nov. 12, 2024**

(30) **Foreign Application Priority Data**

Mar. 1, 2024 (TW) ................................ 113107566

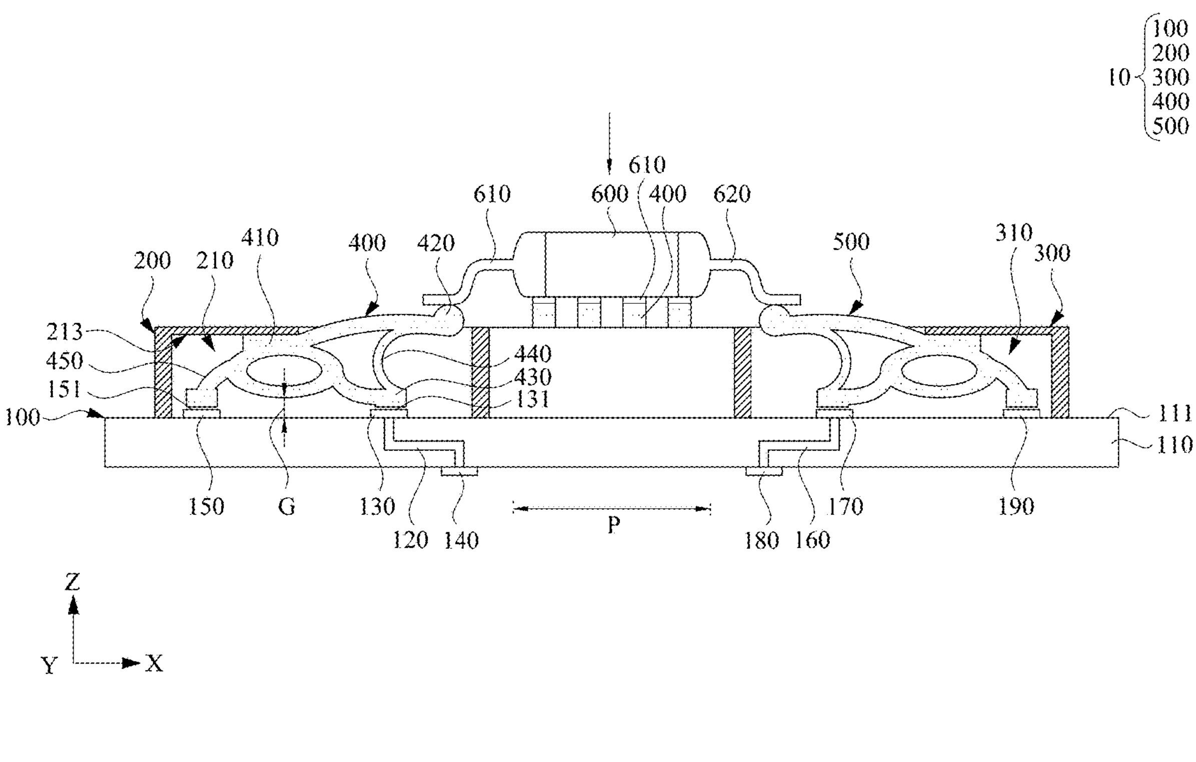
**Publication Classification**

(51) **Int. Cl.**
*G01R 1/04* (2006.01)

(52) **U.S. Cl.**
CPC ................................ ***G01R 1/0416*** (2013.01)

(57) **ABSTRACT**

A testing apparatus includes a circuit board, a base and a conductive terminal. The circuit board includes a board body, a via portion penetrated through the board body, and an electrical contact located on one surface of the board body and electrically connected to the via portion. The base is located on the circuit board and formed with a limiting groove. The conductive terminal includes a main body held within the limiting groove, an abutment portion extended outwardly from one side of the main body, a conductive portion extended outwardly from the side of the main body and removably contacting with the electrical contact, an elastic piece portion connected to the conductive portion and the abutment portion, and a supporting portion extended outwardly from another side of the main body, arranged opposite to the conductive portion, and removably contacting with the circuit board.



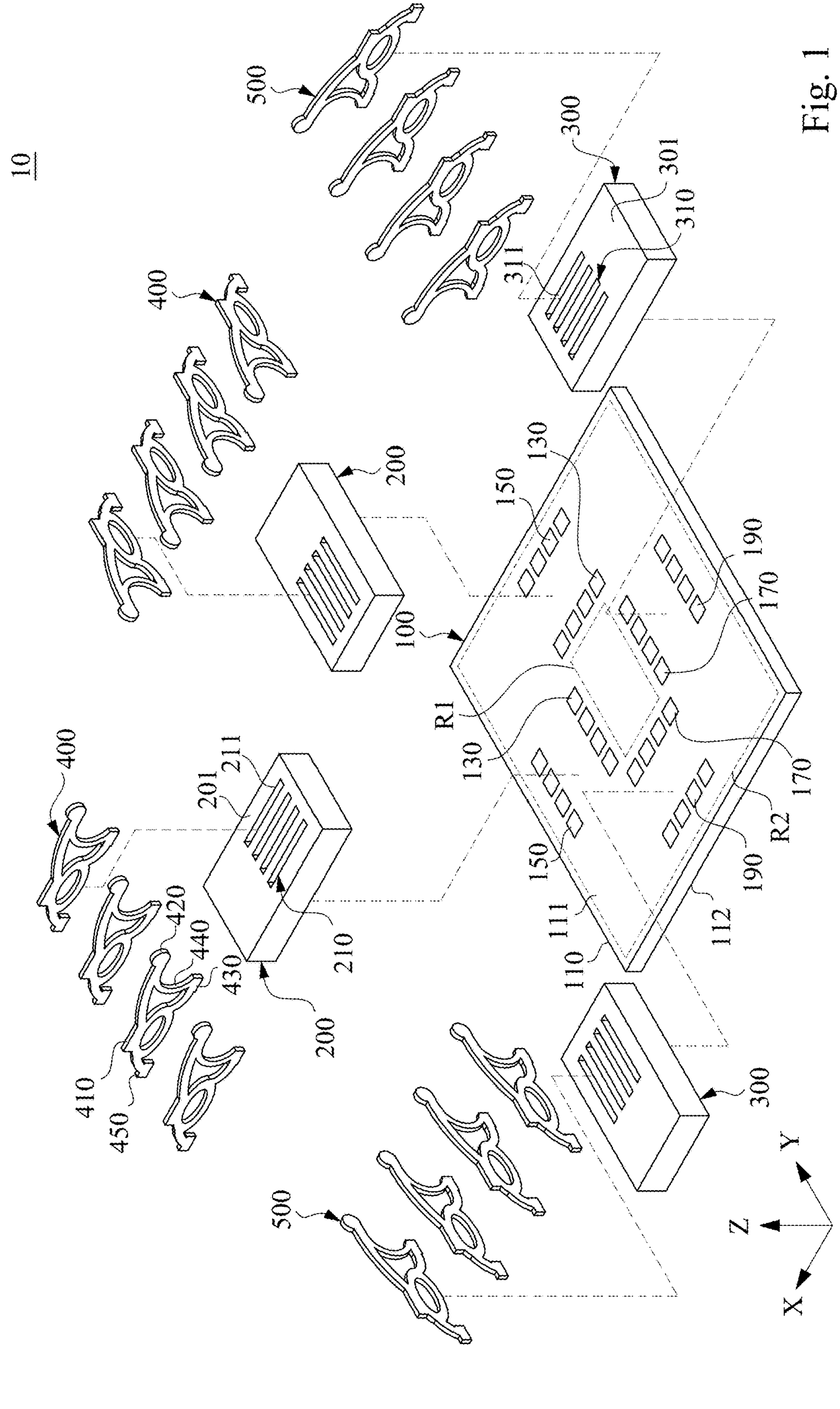

10
500
400
300
301
310
311
200
150
130
190
170
100
R1
130
170
R2
190
150
111
112
110
211
201
400
420
440
430
210
200
410
450
300
500
Z
Y
X

Fig. 1

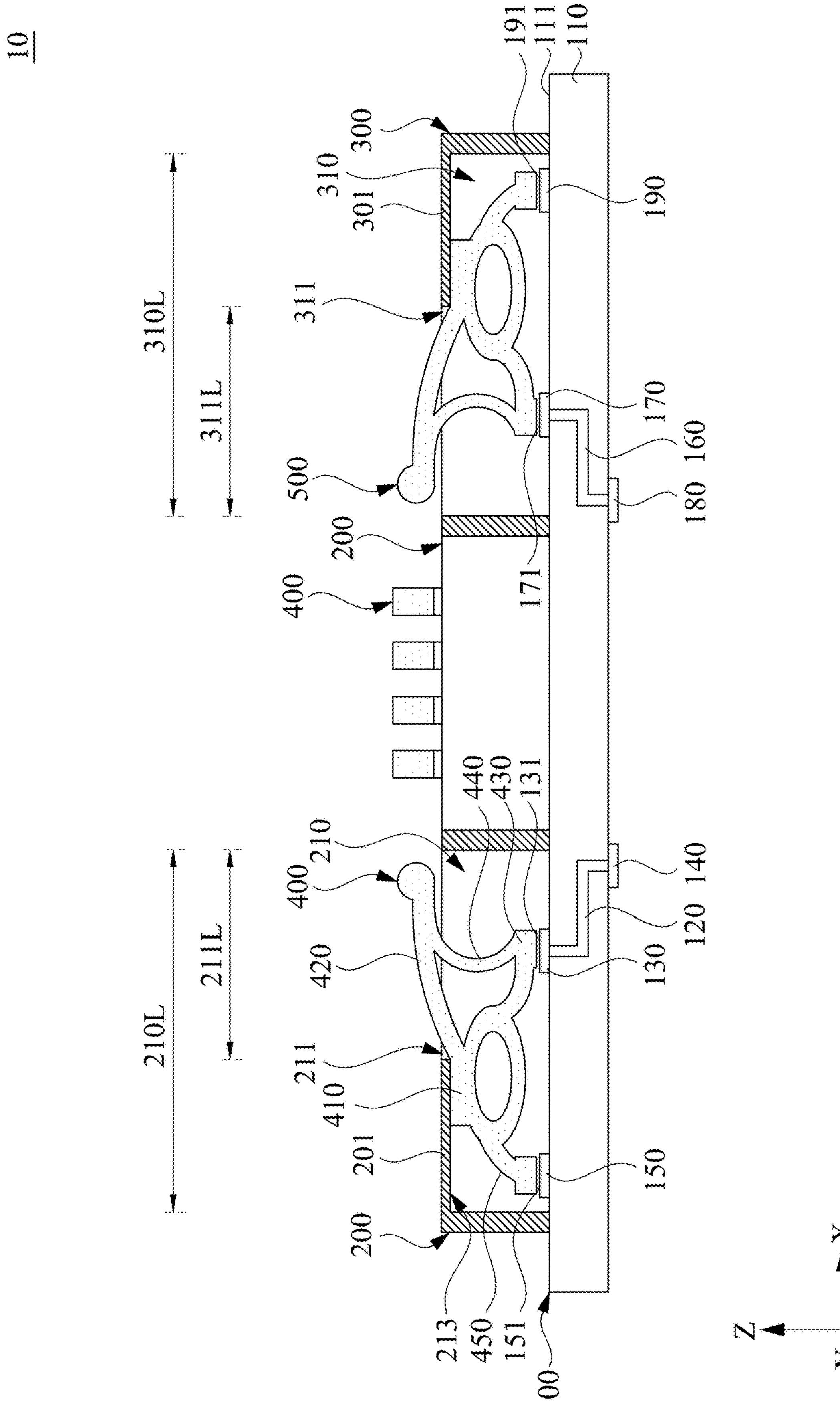

10
191
111
110
300
310
301
190
311
310L
311L
170
160
500
180
200
171
400
440
430
131
210
400
140
211L
420
120
130
210L
211
410
201
150
200
213
450
151
100
Z
X
Y

Fig. 2

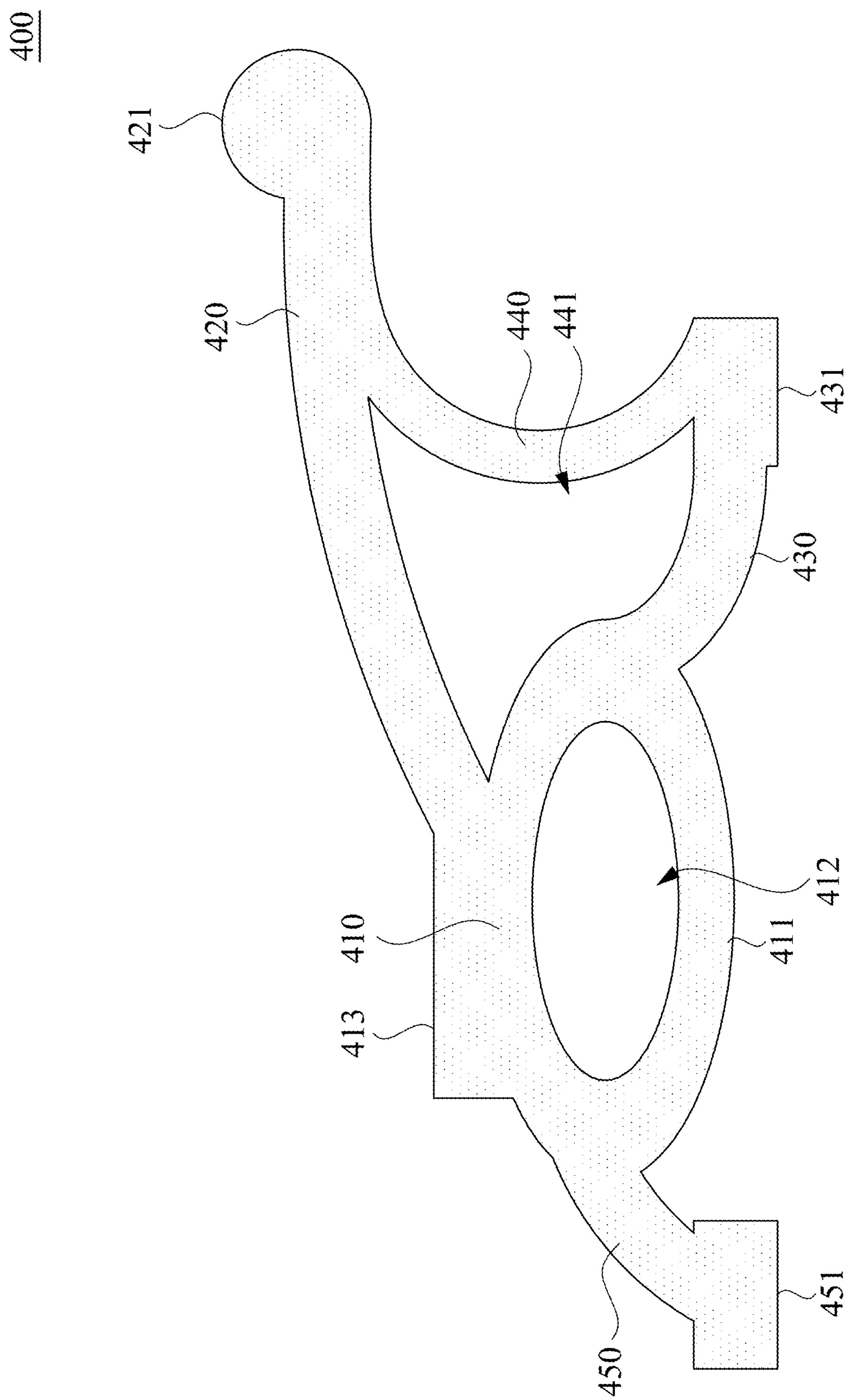



Fig. 3

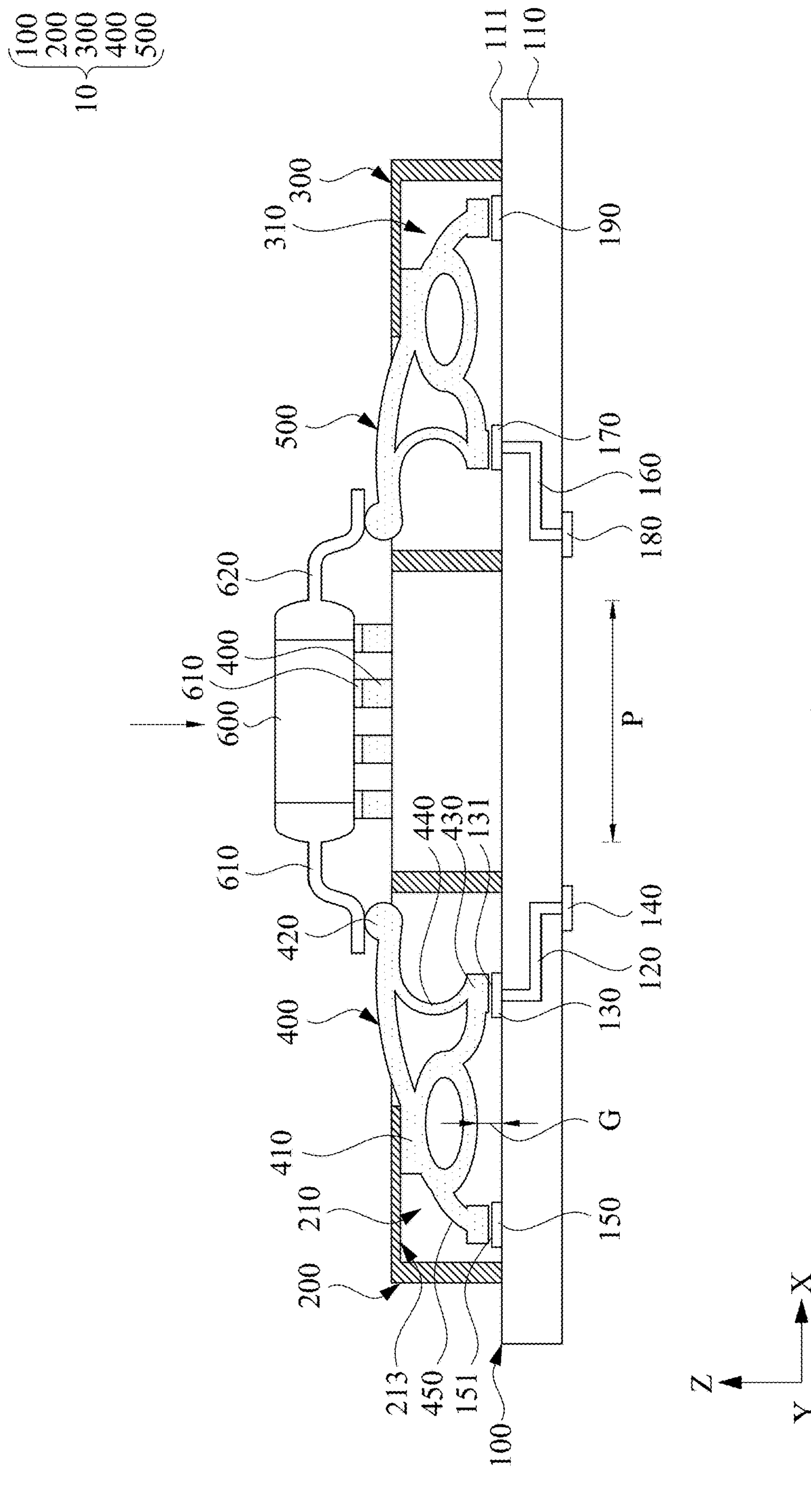

10
100
200
300
400
500
111
110
300
310
190
500
170
160
180
620
610
400
600
610
P
440
430
131
610
420
140
120
130
400
G
410
210
150
200
213
450
151
100
Z
X
Y

Fig. 4

# TESTING APPARATUS AND ITS CONDUCTIVE TERMINAL

## RELATED APPLICATIONS

[0001] This application claims priority to Taiwan Application Serial Number 113107566, filed Mar. 1, 2024, the disclosures of which are incorporated herein by reference in their entireties.

## BACKGROUND

### Field of Invention

[0002] The present disclosure relates to a testing apparatus. More particularly, the present disclosure relates to a testing apparatus of an electronic component and a conductive terminal of the testing apparatus.

### Description of Related Art

[0003] In general, a test operator tests electronic component s through a testing apparatus. The electronic component s are for example, Quad Flat Package (QFP), Low Profile Quad Flat Package (LQFP), Analog to Digital converter (ADC), and digital analog Converter (DAC), etc.

[0004] However, the contact pins of the testing apparatus will cause excessive voltage drop (IR drop) due to their original structural design, which will not only affect the test results of electronic components, but also shorten the service life of contacts on a circuit board of the testing apparatus.

[0005] Therefore, the above-mentioned technologies obviously still have inconveniences and defects, which are issues that the industry needs to solve urgently.

## SUMMARY

[0006] One aspect of the present disclosure is to provide a testing apparatus and its conductive terminal for solving the difficulties mentioned above in the prior art.

[0007] In one embodiment of the present disclosure, a testing apparatus includes a circuit board, at least one base and at least one conductive terminal. The circuit board includes a board body, at least one via portion penetrated through the board body, and at least one electrical contact located on one surface of the board body and electrically connected to the via portion. The base is located on the circuit board and formed with at least one limiting groove. The conductive terminal includes a main body held within the limiting groove, an abutment portion extended outwardly from one side of the main body for contacting with a contact pin of a device under test (DUT), a conductive portion extended outwardly from the side of the main body, and removably contacting with the electrical contact, an elastic piece portion connected to the conductive portion and the abutment portion, and a supporting portion extended outwardly from another side of the main body, arranged opposite to the conductive portion, and removably contacting with the circuit board.

[0008] In one embodiment of the present disclosure, a conductive terminal includes a main body, an abutment portion extended outwardly from one side of the main body for contacting with a contact pin of a device under test (DUT), a conductive portion extended outwardly from the side of the main body for contacting an electrical contact of a circuit board, an elastic piece portion connected to the abutment portion and the conductive portion, and at least one supporting portion extended outwardly from another side of the main body, arranged opposite to the conductive portion, and removably contacting with the circuit board.

[0009] Thus, through the construction of the embodiments above, the disclosure is able to improve the problem of voltage drop caused by the testing apparatus, thus, not only advance the test results of electronic component s, but also maintain the service life of contacts of the circuit board of the testing apparatus.

[0010] The above description is merely used for illustrating the problems to be resolved, the technical methods for resolving the problems and their efficacies, etc. The specific details of the present disclosure will be explained in the embodiments below and related drawings.

[0011] The above description is only an overview of the technical solution of the present disclosure. In order to understand the technical means of the present disclosure, and implement the technical means according to the content of the description, and in order to make the above and other objects, features and advantages of the present disclosure more obvious and understandable, preferred embodiments are specifically cited below and described in detail with reference to the accompanying drawings.

## BRIEF DESCRIPTION OF THE DRAWINGS

[0012] The accompanying drawings are included to provide a further understanding of the present disclosure, and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments of the present disclosure and, together with the description, serve to explain the principles of the present disclosure.

[0013] FIG. **1** is an exploded view of a testing apparatus according to one embodiment of the present disclosure.

[0014] FIG. **2** is a cross-sectional view of the testing apparatus in FIG. **1** after assembly.

[0015] FIG. **3** is an enlargement view of the first conductive terminal in FIG. **1**.

[0016] FIG. **4** is an operative schematic view of the testing apparatus in FIG. **2**.

## DETAILED DESCRIPTION

[0017] Reference will now be made in detail to the present embodiments of the present disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers are used in the drawings and the description to refer to the same or like parts. According to the embodiments, it will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present disclosure without departing from the scope or spirit of the present disclosure.

[0018] Reference is now made to FIG. **1** to FIG. **2**, in which FIG. **1** is an exploded view of a testing apparatus **10** according to one embodiment of the present disclosure. FIG. **2** is a cross-sectional view of the testing apparatus **10** in FIG. **1** after assembly. In this embodiment, the testing apparatus **10** includes a circuit board **100**, a plurality (e.g., four) of bases (refer to a first base **200** and a second base **300**) and a plurality of conductive terminal groups (e.g., four conductive terminal groups). The circuit board **100** includes a board body **110**, a plurality of electrical contacts (e.g., first electrical contacts **130** and second electrical contacts **170** opposite to each other), a plurality of dummy contacts (e.g., first dummy contacts **150** and second dummy contacts **190** opposite to each other), a plurality of rear contacts (e.g., first rear contacts **140** and second rear contacts **180**) and a plurality of via portions (e.g., first via portions **120** and second via portions **160**). The board body **110** includes a top surface **111** and a rear surface **112** opposite to each other. These electrical contacts are spaced located on the top surface **111** of the board body **110**. For example, these electrical contacts (e.g., first electrical contacts **130** and second electrical contacts **170**) are spaced located on the top surface **111** of the board body **110** so as to collectively outline a rectangular contour R1 therein. These dummy contacts are spaced located on the top surface **111** of the board body **110**, and electrically insulated from the via portions and the electrical contacts. For example, these dummy contacts (e.g., these first dummy contacts **150** and these second dummy contacts **190**) are spaced located on the top surface **111** of the board body **110** so as to collectively outline another rectangular contour R2 capable of surrounding these electrical contacts (e.g., first electrical contacts **130** and second electrical contacts **170**). These rear contacts (e.g., first rear contacts **140** and second rear contacts **180**) are spaced located on the rear surface **112** of the board body **110**. The via portions are penetrated through the board body **110**, respectively, and each of the via portions (e.g., first via portion **120**) is electrically connected to one of the electrical contacts (e.g., first electrical contact **130**) and one of the rear contacts (e.g., first rear contacts **140**), respectively.

**[0019]** These bases are mounted on the top surface **111** of the board body **110**. For example, these first bases **200** and these second bases **300** are opposite to each other. Each of the conductive terminal groups is installed in one of the bases, and each of the conductive terminal groups includes a plurality of conductive terminals (refer to first conductive terminals **400** or second conductive terminals **500**). One surface **201** of each of the first bases **200** opposite to the board body **110** is recessed with a plurality of first limiting grooves **210**. These first limiting grooves **210** are arranged linearly in sequence. Each of the first conductive terminals **400** is removably held within one of the first limiting grooves **210**, and each of the first conductive terminals **400** located within one of the first bases **200** is removably placed against (or at least contacted with) the first electrical contact **130** and the circuit board **100** (e.g., first dummy contact **150**), respectively. The opening length **211**L of a recess opening **211** of each of the first limiting grooves **210** is smaller than the inner length **210**L of the first limiting groove **210**. One of the second bases **300** is opposite to one of the first bases **200**. One surface **301** of each of the second base **300** opposite to the board body **110** is recessed with a plurality of second limiting grooves **310**. These second limiting grooves **310** are arranged linearly in sequence. Each of the second conductive terminals **500** is removably held within one of the second limiting grooves **310**, and each of the second conductive terminals **500** located within one of the second bases **300** is removably placed against (or at least contacted with) the second electrical contact **170** and the circuit board **100** (e.g., second dummy contact **190**), respectively.

**[0020]** The opening length **311**L of a recess opening **311** of each of the second limiting grooves **310** is smaller than the inner length **310**L of the second limiting groove **310**. The arrangement direction of each of the second limiting grooves **310** (e.g., Y axis) is parallel to the arrangement direction of each of the first limiting grooves **210** (e.g., Y axis).

**[0021]** FIG. **3** is an enlargement view of the first conductive terminal **400** in FIG. **1**. As shown in FIG. **2** and FIG. **3**, in the embodiment, each of the conductive terminals (e.g., first conductive terminal **400**) includes a main body **410**, an abutment portion **420**, a conductive portion **430**, an elastic piece portion **440** and a supporting portion **450**. The main body **410** is removably held within the first limiting groove **210**. The abutment portion **420** is connected to one side of the main body **410**, and extended outwards the first limiting groove **210** from the main body **410**. The conductive portion **430** is connected to the main body **410**, and extended outwardly from the aforementioned side of the main body **410**. The conductive portion **430** removably abuts or at least contacts with the first electrical contact **130**. The elastic piece portion **440** is resiliently located between the abutment portion **420** and the conductive portion **430**, and connected to the abutment portion **420** and the conductive portion **430**, respectively. The supporting portion **450** is connected to the main body **410**, and extended from another side of the main body **410**. The supporting portion **450** is arranged opposite to the conductive portion **430**, and removably abutted or at least contacted with the circuit board **100**.

**[0022]** FIG. **4** is an operative schematic view of the testing apparatus **10** in FIG. **2**. As shown in FIG. **3** and FIG. **4**, when a device under test (called DUT hereinafter, e.g., an electronic component) is placed on the testing apparatus **10**, since contact pins (e.g., contact pins **610** and **620**) on each side of the DUT **600** are arranged at intervals according to a rectangular contour, the contact pins **610** and **620** of the DUT **600** respectively contact with the first conductive terminals **400** and the second conductive terminals **500** one by one. The abutment portions **420** of each of the first conductive terminals **400** and the second conductive terminals **500** are respectively placed against (or at least contacted with) one of the contact pins **610** and **620** of the DUT **600**, so that the DUT **600** and the testing apparatus **10** are electrically connected to each other.

**[0023]** In this way, when the DUT **600** presses down these conductive terminals (e.g., the first conductive terminals **400** and the second conductive terminals **500**), since the elastic piece portion **440** of the first conductive terminals **400** can provide a certain amount of elastic deformation, the pressure exerted by the conductive portions **430** of the first conductive terminals **400** on the first electrical contacts **130** can be buffered. Similarly, since the corresponding position of the second conductive terminals **500** can also provide a certain amount of elastic deformation, the pressure exerted by the second conductive terminal **500** on the second electrical contacts **170** can also be buffered. In this way, it not only stabilizes the electrical qualities of the first conductive terminals **400** and the second conductive terminals **500** to the first electrical contacts **130** and second electrical contacts **170**, respectively, but also improves the problem of possible voltage drop in the testing apparatus **10**. Therefore, the test results of the DUT **600** are improved and the service life of the circuit board **100** contacts of the testing apparatus **10** is maintained.

**[0024]** More specifically, in this embodiment, as shown in FIG. **3** and FIG. **4**, the conductive portion **430** of each of the first conductive terminals **400** is a cantilever extending towards the second bases **300** from the main body **410** thereof. The supporting portion **450** of each of the first conductive terminals **400** is a cantilever extending from the main body **410** in a direction away from the abutment portion **420**. The abutment portion **420** of each of the first conductive terminals **400** is a cantilever extending laterally toward the second base **300** from the main body **410** thereof. One end of the abutment portion **420** facing away from the main body **410** is provided with an arc-shaped outer edge **421**. When the DUT **600** presses the end of the abutment portion **420** facing away from the main body **410**, the abutment portion **420** is in contact with the contact pins **610** of the DUT **600** through the arc-shaped outer edge **421**, thereby testing residue is not easy to be accumulated on the end of the abutment portion **420**.

**[0025]** However, the present disclosure is not limited thereto, in other embodiments, the supporting portion **450** may include two supporting portions **450**, and these supporting portions **450** and the conductive portion **430** are arranged in a triangle together.

**[0026]** Furthermore, the main body **410** of each of the first conductive terminals **400** is recessed with a recess **441** connected to the elastic piece portion **440** thereof. The elastic piece portion **440** is convex to protrude into the recess **441** so that a geometric structure formed by the elastic piece portion **440** and the recess **441** can provide a suitable elastic coefficient. Therefore, when the DUT **600** presses down one of the first conductive terminals **400**, the recess **441** of the corresponding first conductive terminal **400** allows the elastic piece portion **440** of the corresponding first conductive terminal **400** to continue extending inwards the recess **441**, thereby enhancing the elastic deformation of the elastic piece portion **440** and the stability of the first conductive terminal **400**. For example, the elastic piece portion **440** is C-shaped or V-shaped, however, the present disclosure is not limited thereto.

**[0027]** Furthermore, as shown in FIG. **2**, one surface **151** of the first dummy contact **150** and one surface **131** of the first electrical contact **130** which are connected to the same first conductive terminal **400**, are at the same height. In this way, when each of the first conductive terminals **400** is placed on the circuit board **100**, the conductive portion **430** of the first conductive terminal **400** is placed against (or at least contacted with) the aforementioned surface **131** of the corresponding first electrical contact **130**, and the supporting portion **450** is placed against (or at least contacted with) the aforementioned surface **151** of the corresponding first dummy contact **150**. Similarly, one surface **191** of the second dummy contact **190** and one surface **171** of the second electrical contacts **170** which are connected to the same second conductive terminal **500** are at the same height.

**[0028]** Furthermore, for example, the conductive portion **430** of each of the first conductive terminals **400** is a cantilever extending towards the second bases **300** from the main body **410** thereof, the conductive portion **430** is provided with a conductive surface **431** located away from the main body **410** and the elastic piece portion **440**. The conductive surface **431** (FIG. **3**) is used to be placed against (or at least contact with) the surface **131** of the corresponding first electrical contact **130**. The supporting portion **450** is a cantilever extending in the direction away from the abutment portion **420** from the main body **410**, one end surface of the supporting portion **450** away from the main body **410** is provided with a supporting surface **451**. The supporting surface **451** is used to be placed against (or at least contact with) the surface **151** of the corresponding first dummy contact **150**.

**[0029]** Thus, since the aforementioned surface **151** of the first dummy contact **150** is at the same height as the surface **131** of the corresponding first electrical contact **130**, the corresponding first conductive terminal **400** can be tested at the same level, so that the first conductive terminals **400** can be more stable test the DUT **600**.

**[0030]** Furthermore, since the hardness of one of the first electrical contacts **130**, the second electrical contacts **170**, the first dummy contacts **150** and the second dummy contacts **190** is greater than the hardness of the board body **110** of the circuit board **100**, the supporting portion **450** of the first conductive terminal **400** abuts against the first dummy contact **150** rather than the board body **110** of the circuit board **100**. In this way, the supporting portion **450** of each of the first conductive terminals **400** can avoid excessive wear of the board body **110** of the circuit board **100** over a long period of time. In this embodiment, the board body **110** is made of plastic, for example, and the first electrical contacts **130**, the second electrical contacts **170**, the first dummy contacts **150** and the second dummy contacts **190** are made of metal, for example.

**[0031]** However, the present disclosure is not limited thereto. In other embodiments, the supporting portion **450** of each of the first conductive terminals **400** and each of the second conductive terminals **500** may also be placed against (or at least contact with) the board body **110** of the circuit board **100**, rather than the electrical contacts.

**[0032]** In this embodiment, as shown in FIG. **4**, comparing to the first dummy contacts **150**, the first electrical contact **130** of the circuit board **100** is closer to an orthographic projection P of the DUT **600** onto the board body **110** or the second electrical contact **170** of the circuit board **100**, so that signals of the DUT **600** can be sent to the circuit board **100** from the abutment portion **420** via the elastic piece portion **440** and the conductive portion **430** through the minimum resistance and path.

**[0033]** As shown in FIG. **3** and FIG. **4**, each of the first conductive terminals **400** is further provided with an elastic ring **411** located between the conductive portion **430** and the supporting portion **450**. The elastic ring **411** surrounds to form a through hole **412** therein so that the elastic ring **411** is collectively supported by the conductive portion **430** and the supporting portion **450** in a suspended manner, thereby maintaining a gap G between the elastic ring **411** and the top surface **111** of the board body **110** of the circuit board **100** so as to form a suspension mechanism capable of absorbing vibrations. However, the present disclosure is not limited thereto, the elastic ring **411** may also be in contact with the top surface **111** of the board body **110** in other requirements.

**[0034]** Each of the first conductive terminals **400** is further provided with a pre-press portion **413**. The pre-press portion **413** is opposite to the elastic ring **411** (or the board body **110** of the circuit board **100**), and arranged between the abutment portion **420** and the supporting portion **450** for abutting against the inner wall **213** of the first base **200** so that the first base **200** exerts a preload pressure on the first conductive terminal **400**, thereby pressing the conductive terminal on the board body **110** of the circuit board **100**. Therefore, the first conductive terminal **400** can be more stably abutted against the first electrical contact **130** and the first dummy contact **150**, thereby more stabilizing the electrical quality of the first conductive terminal **400** to the first electrical contact **130**.

**[0035]** In addition, as shown in FIG. **4**, when the DUT **600** presses down the first conductive terminal **400** and the second conductive terminal **500**, since the first conductive terminal **400** directly abuts against the inner wall **213** of the first base **200** on the one hand, the first conductive terminal **400** directly abuts the first electrical contact **130** and the first dummy contact **150** on the other hand, not only allows the first conductive terminals **400** to stably abut against the first electrical contacts **130** and the first dummy contacts **150**, but also prevents the first conductive terminals **400** from generating rolling friction based on the conductive portion **430** as a fulcrum, thereby reducing the wear damages of the first electrical contacts **130**.

**[0036]** In the embodiment, the abutment portion **420**, the conductive portion **430**, the elastic piece portion **440** and the supporting portion **450** are integrally connected to the main body **410**. More particularly, the abutment portion **420**, the conductive portion **430**, the elastic piece portion **440**, the elastic ring **411**, the pre-press portion **413** and the supporting portion **450** are integrally connected to the main body **410**, however, the disclosure is not limited thereto. The aforementioned electronic component for example, can be one of a Quad Flat Package (QFP), a Low Profile Quad Flat Package (LQFP), an Analog to Digital converter (ADC), and a digital analog Converter (DAC). However, the disclosure is not limited thereto.

**[0037]** Thus, through the construction of the embodiments above, the disclosure is able to improve the problem of voltage drop caused by the testing apparatus, thus, not only advance the test results of electronic components, but also maintain the service life of contacts of the circuit board of the testing apparatus.

**[0038]** Although the present invention has been described in considerable detail with reference to certain embodiments thereof, other embodiments are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein.

**[0039]** It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present invention without departing from the scope or spirit of the invention. In view of the foregoing, it is intended that the present invention cover modifications and variations of this invention provided they fall within the scope of the following claims.

What is claimed is:

**1**. A testing apparatus, comprising:

a circuit board comprising a board body, at least one via portion penetrated through the board body, and at least one electrical contact located on one surface of the board body and electrically connected to the at least one via portion;

at least one base located on the circuit board and formed with at least one limiting groove; and

at least one conductive terminal comprising:

a main body held within the at least one limiting groove;

an abutment portion extended outwardly from one side of the main body for contacting with a contact pin of a device under test (DUT);

a conductive portion extended outwardly from the side of the main body, and removably contacting with the at least one electrical contact;

an elastic piece portion connected to the conductive portion and the abutment portion; and

at least one supporting portion extended outwardly from another side of the main body, arranged opposite to the conductive portion, and removably contacting with the circuit board.

**2**. The testing apparatus of claim **1**, wherein the circuit board further comprises:

at least one dummy contact located on the surface of the board body, electrically insulated from the at least one via portion and the at least one electrical contact, and one surface of the at least one dummy contact and one surface of the at least one electrical contact are at the same height,

wherein the conductive portion is in contact with the surface of the at least one electrical contact, and the at least one supporting portion is in contact with the surface of the at least one dummy contact.

**3**. The testing apparatus of claim **2**, wherein the at least one electrical contact of the circuit board is closer to an orthographic projection of the DUT onto the board body than the at least one dummy contact.

**4**. The testing apparatus of claim **1**, wherein the main body is recessed with a recess connected to the elastic piece portion, and the elastic piece portion protrudes toward the recess.

**5**. The testing apparatus of claim **1**, wherein the elastic piece portion is C-shaped or V-shaped.

**6**. The testing apparatus of claim **1**, wherein the abutment portion is provided with an arc-shaped outer edge, and the arc-shaped outer edge is used to be in contact with the contact pin.

**7**. The testing apparatus of claim **1**, wherein the main body is further provided with an elastic ring, the elastic ring is formed with a through hole, located between the conductive portion and the at least one supporting portion, and collectively supported by the conductive portion and the at least one supporting portion in a suspended manner.

**8**. The testing apparatus of claim **1**, wherein the main body is further provided with a pre-press portion that is opposite to the board body and between the abutment portion and the at least one supporting portion for abutting against the base so that the at least one conductive terminal is pressed to the circuit board.

**9**. The testing apparatus of claim **1**, wherein the abutment portion, the conductive portion, the elastic piece portion and the at least one supporting portion are integrally connected to the main body.

**10**. The testing apparatus of claim **1**, wherein the at least one supporting portion comprises two supporting portions, and the supporting portions and the conductive portion are arranged in a triangle together.

**11**. The testing apparatus of claim **2**, wherein the at least one electrical contact comprises a plurality of electrical contacts spaced arranged on the surface of the board body so as to collectively outline a first rectangular contour;

the at least one dummy contact comprises a plurality of dummy contact spaced arranged on the surface of the board body so as to collectively outline a second rectangular contour surrounding the electrical contacts therein; and the at least one base comprises four bases spaced arranged on the surface of the board body according to the first rectangular contour,

wherein the testing apparatus is used to test an electronic component which is one of a Quad Flat Package (QFP), a Low Profile Quad Flat Package (LQFP), an Analog to Digital converter (ADC), and a digital analog Converter (DAC).

**12**. A conductive terminal, comprising:

a main body;

an abutment portion extended outwardly from one side of the main body for contacting with a contact pin of a device under test (DUT);

a conductive portion extended outwardly from the side of the main body for contacting an electrical contact of a circuit board;

an elastic piece portion connected to the abutment portion and the conductive portion; and

at least one supporting portion extended outwardly from another side of the main body, arranged opposite to the conductive portion, and removably contacting with the circuit board.

**13**. The conductive terminal of claim **12**, wherein the main body is recessed with a recess connected to the elastic piece portion, and the elastic piece portion protrudes toward the recess.

**14**. The conductive terminal of claim **12**, wherein the elastic piece portion is C-shaped or V-shaped.

**15**. The conductive terminal of claim **12**, wherein the abutment portion is provided with an arc-shaped outer edge, and the arc-shaped outer edge is used to be in contact with the contact pin.

**16**. The conductive terminal of claim **12**, wherein the main body is further provided with an elastic ring, the elastic ring is formed with a through hole, located between the conductive portion and the at least one supporting portion, and collectively supported by the conductive portion and the at least one supporting portion in a suspended manner.

**17**. The conductive terminal of claim **12**, wherein the main body is further provided with a pre-press portion that is located between the abutment portion and the at least one supporting portion.

**18**. The conductive terminal of claim **12**, wherein the abutment portion, the conductive portion, the elastic piece portion and the at least one supporting portion are integrally connected to the main body.

**19**. The conductive terminal of claim **12**, wherein the at least one supporting portion comprises two supporting portions, and the supporting portions and the conductive portion are arranged in a triangle together.

* * * * *